WORKED SOLUTION

STEP I 2000, Question 11

Mechanics • Ladder Friction Equilibrium

STEP I 2000, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q11

The key idea

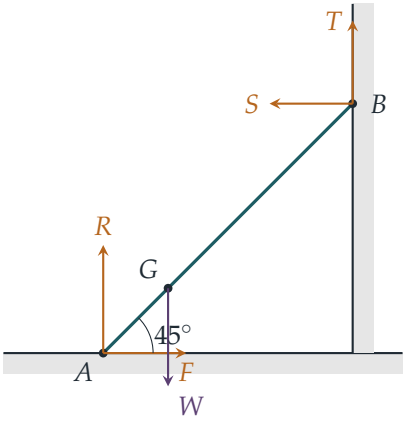
This is a ladder problem with friction at both ends. "Limiting at both A and B" fixes every frictional force in terms of a normal reaction, so the two resolving equations determine all four contact forces. The moment equation is then one condition too many: either it happens to hold, or it fails. For the first part we show that it fails. For the second part we use the position of the extra mass as the unknown that makes it hold.

Solution

The model and the equations

We model the rod as a rigid rod with its weight $W = 5g\,\mathrm{N}$ acting at G. The contacts at A (floor) and B (wall) are both rough. The forces are:

- at A: normal reaction R upwards and friction F horizontally;
- at B: normal reaction S horizontally away from the wall and friction T vertically (taken as upwards, so $T < 0$ would mean downwards);
- the weight W downwards at G.



The only horizontal forces are F and S, and $S \geq 0$ pushes the rod away from the wall. So F must act towards the wall, with

$$F = S. \tag{1}$$

Resolving vertically,

$$R + T = W. \tag{2}$$

For moments about A, put $AB = 0.81$ and $c = \cos 45^\circ = \sin 45^\circ$. Then B is a horizontal distance $0.81c$ and a height $0.81c$ from A, and G is a horizontal distance $0.21c$ from A. Taking clockwise and anticlockwise moments,

$$S\,(0.81c) + T\,(0.81c) = W\,(0.21c),$$

and cancelling c gives

$$0.81\,(S + T) = 0.21\,W. \tag{3}$$

This is where the 45° helps: the factor c cancels from every term.

Friction cannot be limiting at both ends

Suppose friction is limiting at both ends. Then $F = 0.2R$ and $|T| = 1.0\,S = S$.

Ask yourself Which way does the friction at B act when it is limiting?

The natural case is that the rod is about to slip with A moving away from the wall and B sliding down, so that $T = +S$. To be thorough we check both signs.

Case $T = S$. From (1), $S = F = 0.2R$, and (2) gives $R + 0.2R = W$. So $R = \frac{5}{6}W$ and $S = T = \frac{1}{6}W$. Then the left-hand side of (3) is

$$0.81\left(\tfrac{1}{6}W + \tfrac{1}{6}W\right) = 0.27\,W \neq 0.21\,W.$$

So the moment equation fails.

Case $T = -S$. The left-hand side of (3) is 0, but the right-hand side is $0.21W > 0$, which is impossible.

In both cases the rod could not be in equilibrium. So the friction cannot be limiting at both A and B, as required.

(Equilibrium is certainly possible in the non-limiting case: for instance $T = S = \frac{7}{54}W$ satisfies (3) and gives $R = \frac{47}{54}W$, with $F = S \leq 0.2R$.)

Adding the mass at P

Let $AP = x$ m. The extra weight is also $W = 5g$ N, acting at P, so the total weight is $2W$. Equations (1) and (2) become

$$F = S, \qquad R + T = 2W,$$

and the moment equation about A gains a term $W(xc)$. After cancelling c it reads

$$0.81\,(S + T) = 0.21\,W + x\,W. \tag{4}$$

Now friction is limiting at both ends. The case $T = -S$ is again impossible, because then the left-hand side of (4) is 0 but the right-hand side is positive. So $T = S$ and $F = 0.2R$, and therefore

$$S = 0.2R, \qquad R + 0.2R = 2W \quad \Longrightarrow \quad R = \tfrac{5}{3}W, \quad S = T = \tfrac{1}{3}W.$$

Substituting into (4) gives

$$0.81 \times \tfrac{2}{3}\,W = 0.54\,W = 0.21\,W + x\,W,$$

so $x = 0.33$. Hence

$$AP = 0.33 \text{ m}.$$

This lies between 0 and 0.81, so P is on the rod. It is beyond G (and beyond the midpoint), so the extra mass pushes the combined centre of gravity up the rod, and that tends to make a ladder slip.

Checks

- With the mass at P, the forces in units of W are $R = \frac{5}{3}$ and $F = S = T = \frac{1}{3}$. Then $F/R = 0.2$ and $T/S = 1$, so both are limiting. Vertically $\frac{5}{3} + \frac{1}{3} = 2$, and the moments give $0.81 \times \frac{2}{3} = 0.54 = 0.21 + 0.33$.
- The combined centre of gravity is at $\frac{1}{2}(0.21 + 0.33) = 0.27$ m from A. That is exactly the position the first part needed for limiting friction at both ends ($0.27W$ against $0.21W$), which makes sense because the coefficients are the same and only the total weight has doubled.
- The mass and g cancel throughout, so the answer is independent of them. A numerical substitution with $g = 9.8$ and the true 45° moments leaves residuals of order 10^{-15}.

Common slips

- Taking the friction at A in the wrong direction. Since S is the only other horizontal force, F must point towards the wall.
- Forgetting that the friction at B is vertical, so it appears in the vertical resolution and reduces R below the total weight.
- Using the midpoint of the rod instead of G. The rod is not uniform.

Beyond STEP

A rigid body in a plane gives three equilibrium equations, so a problem with four unknown contact forces is *statically indeterminate*. Rigid-body statics alone cannot tell you how the friction is shared between the ends. Imposing "limiting at both ends" adds two more equations, which over-determines the system, and that is exactly why the first part has no solution. In engineering, indeterminacy is resolved by modelling the elasticity of the contacts. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully at first-year level.

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